D-UNIQE: DESCRIPTIVE UNIFIED SPEECH QUALITY AND EMOTION ASSESSMENT FOR EMOTIONAL TTS

Jaeyoung Shin[⋆] *Kyogu Lee*[⋆†]

[⋆] Music and Audio Research Group (MARG),
Interdisciplinary Program in Artificial Intelligence, Seoul National University
[†] Department of Intelligence and Information, Seoul National University

ABSTRACT

Text-to-Speech (TTS) systems have achieved substantial improvements in naturalness and emotional expressiveness. However, evaluation of TTS still relies on human annotation, and objective metrics lack interpretability as they provide only scalar outputs. Recent audio large language models can generate scores and explanations, but often produce inaccurate or inconsistent responses. To address this, we propose a Descriptive Unified Quality evaluation framework for Emotional TTS that jointly models speech quality and emotional fidelity within a single model. Our approach integrates semantic and acoustic encoders with continuous emotion representations and speaker-attribute conditioning to better capture perceptual speech features. A Task Routing Module mitigates multi-task interference, and GRPO-QTR (Group-Relative Policy Optimization with Quality-Trust Reasoning) training improves alignment between scores and explanations. Experiments show that our approach yields more reliable scores and coherent rationales than prior speech LLMs. Samples are available at `https://jyshin0926.github.io/DUNIQE/`.

***Index Terms*—** speech quality assessment, speech emotion recognition, automatic metrics, speech large language model

1. INTRODUCTION

Text-to-Speech (TTS) systems have recently achieved remarkable gains in perceived naturalness, accelerating progress toward expressive and emotionally controllable speech generation. Despite these advances, TTS evaluation still relies heavily on human listening tests, most commonly Mean Opinion Score (MOS), as the gold standard for assessing speech quality. As TTS research scales, there is a growing need for automatic quality assessment methods that are cost-effective, faithful to human perception, and useful for debugging.

Beyond naturalness, modern systems must be assessed for emotional fidelity: whether the generated speech matches a target emotion and how convincingly that emotion is conveyed. A straightforward solution is to maintain separate evaluators for speech quality assessment (SQA) and speech emotion recognition (SER), but this increases engineering overhead and yields fragmented feedback. Conversely, naively unifying multiple perceptual tasks into a single model often degrades performance due to conflicting objectives and shared representations. This motivates a unified evaluator that can reliably handle both naturalness scoring and emotion assessment within one framework.

Recent SQA work has explored audio Large Language Models (LLMs) as perceptual evaluators that can predict scores and generate textual rationales. However, current models often struggle with *prompt–response accuracy* and exhibit *answer–reason inconsistency*, where fluent explanations do not align with the predicted score. Notably, this inconsistency frequently persists even when generating the answer and rationale separately, suggesting that the evaluator lacks robustness. Moreover, while answer–reason consistency has been studied primarily in mathematical problem solving, it remains underexplored in speech evaluation, where rationales must be grounded in perceptual cues rather than symbolic steps. In addition, many prior approaches depend on new datasets with extra human annotations, limiting scalability.

In this paper, we propose a unified descriptive evaluation framework for expressive TTS with three objectives: (1) **multi-task evaluation** of naturalness, audio quality and emotion within a single model, (2) **perceptual explanations aligned with ground-truth labels** without additional annotation, and (3) **consistent answer–reason generation** for interpretable judgments.

To obtain perceptually grounded representations, we adopt a *disentangled multi-encoder* design comprising a semantic encoder for speech content, an acoustic encoder for general audio events, and an emotion encoder that predicts arousal, valence, and dominance directly from audio, reducing reliance on lexical content when estimating emotional expression. These complementary representations are fused and projected into token-level inputs for the LLM via a Q-Former modality adapter. To mitigate performance degradation caused by conflicting objectives in joint SQA and SER modeling, we introduce a *task-routed projection mixture-of-experts* that dynamically combines task-specific experts conditioned on the assessment task, allowing the model to activate distinct perceptual cues for naturalness and audio quality scoring versus emotion assessment. Finally, we apply *GRPO-QTR* (Group-Relative Policy Optimization with Quality-Trust Reasoning) with a bounded score-accuracy reward and a trust-gated quality-reasoning reward to improve both numerical accuracy and alignment between the predicted score and the generated explanation on SQA tasks.

2. RELATED WORKS

Speech LLMs Speech LLMs such as SALMONN [1] enable audio-conditioned instruction following, motivating LLM-based evaluators that can produce natural-language justifications [2, 3]. However, prompt-only scoring is often unreliable for MOS-style prediction, which the model may output incorrect scores or produce rationales that do not match the score, and outputs can be sensitive to prompt phrasing. We therefore target **score reliability** and **answer–reason consistency** rather than relying on prompting alone.

Speech quality assessment. Classic metrics such as PESQ [4] and POLQA [5] focus on limited degradation types and do not reflect

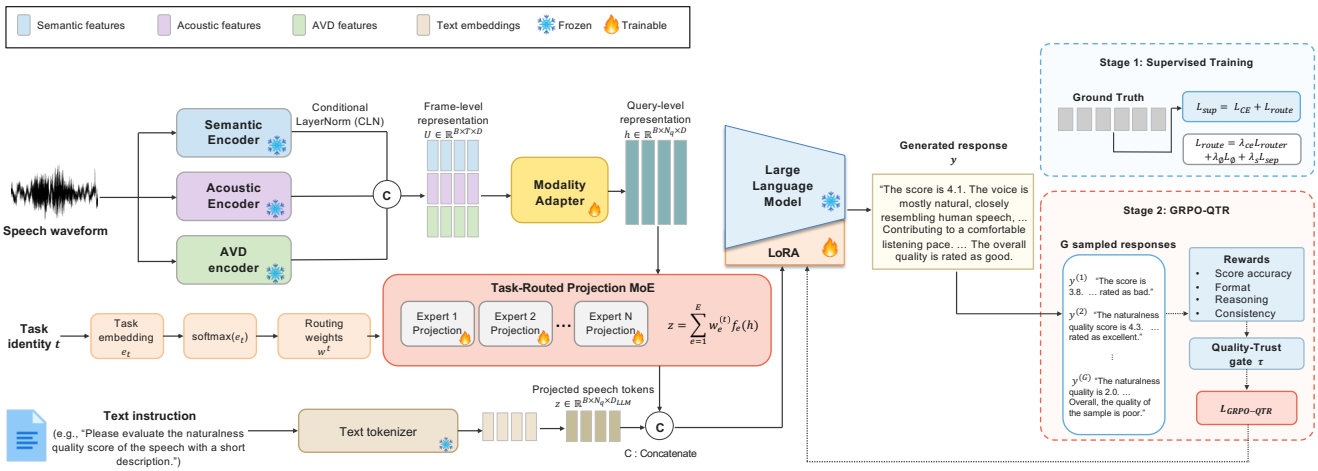


Fig. 1. Overview of the proposed descriptive speech quality assessment framework.

TTS naturalness and expressiveness. Learning-based metrics [6, 7] improve correlation with human judgments but provide limited interpretability. QualiSpeech [8] and recent descriptive-evaluator work [2] move toward explanation-based assessment. However, prior methods rarely address emotional TTS with a unified evaluator for naturalness and emotion accuracy, nor do they explicitly enforce consistency between scores and explanations. Our framework unifies these tasks and improves robustness through consistency-aware training.

Speech emotion recognition. Speech Emotion Recognition (SER) task has been studied with discrete labels and dimensional affect [9]. SSL or Transformer approaches improve Arousal-Valence-Dominance (AVD) prediction [10, 9], emotion2vec [11] offers strong emotion representations, and LLM-based SER systems [12, 13] improve categorical recognition. In this work, we incorporate an AVD-based emotion encoder as explicit emotional evidence grounded in the audio signal rather than relying on text alone, enabling the model to assess whether synthesized speech matches the target emotion.

3. METHOD

D-UNIQE combines a frozen multi-encoder speech front-end with a windowed Q-Former and a LoRA-adapted Vicuna-7B decoder, supporting three assessment tasks: naturalness, audio quality, and emotion. We introduce two main components: (i) a task-routed projection MoE that learns task-specific speech-to-LLM mappings while retaining soft parameter sharing, and (ii) *GRPO-QTR* (Group-Relative Policy Optimization with Quality-Trust Reasoning), which improves continuous score accuracy and score–explanation consistency on SQA tasks.

3.1. Multi-Encoder Speech Representation

The frozen Whisper [14], BEATs [15], and AVD [9] encoders extract semantic, acoustic, and affective features, respectively, where the AVD branch is used only for emotion assessment. Whisper features are conditioned on precomputed ECAPA-TDNN [16] speaker embeddings through conditional layer normalization. The resulting features are temporally aligned and fused into $\mathbf{U} \in \mathbb{R}^{B \times T_a \times D}$.

A windowed Q-Former [17] compresses the frame-level sequence within local temporal windows:

$$\mathbf{Z}_w = \mathrm{softmax}\left(\frac{(\mathbf{Q}\mathbf{W}_Q)(\mathbf{U}_w\mathbf{W}_K)^\top}{\sqrt{d_k}}\right)(\mathbf{U}_w\mathbf{W}_V), \quad (1)$$

where $\mathbf{Q}$ denotes trainable query tokens. The window-level outputs are concatenated and projected into the LLM embedding space by the task-routed module.

3.2. Task-Routed Projection MoE

A single shared projection may cause negative transfer across tasks with different perceptual cues. We use E=3 linear projection experts whose routing weights depend on the assessment *task identity*. For task t, a learned embedding $\mathbf{e}_t \in \mathbb{R}^E$ (zero-initialized) produces

$$\mathbf{w}^{(t)} = \mathrm{softmax}(\mathbf{e}_t), \quad (2)$$

and the projected representation is $\mathbf{z} = \sum_e w_e^{(t)} f_e(\mathbf{h})$, where $\mathbf{h}$ is the Q-Former output and all experts are mixed densely.

To promote specialization while retaining soft sharing, we apply

$$\mathcal{L}_{\text{route}} = \lambda_{\text{ce}}\mathcal{L}_{\text{router}} + \lambda_\phi\mathcal{L}_\phi + \lambda_s\mathcal{L}_{\text{sep}}. \quad (3)$$

$\mathcal{L}_{\text{router}}$ is a cross-entropy that anchors task i to expert i; $\mathcal{L}_\phi$ is a task-scoped ϕ-balancing [18] term that pulls each task's routing load toward a task-specific prior; and $\mathcal{L}_{\text{sep}}$ is the negative pairwise Jensen–Shannon divergence between task routing distributions, encouraging different tasks to adopt distinct expert mixtures.

3.3. GRPO-QTR

Supervised training does not directly optimize continuous score accuracy or score–explanation consistency. GRPO-QTR is applied only to score-based SQA tasks (naturalness and audio quality); emotion batches remain supervised cross-entropy. For each SQA input x, we sample G responses and assign

$$R^{(g)} = \alpha_o R^{(g)}_{\text{out}} + \alpha_f R^{(g)}_{\text{fmt}} + \tau\lambda_r R^{(g)}_{\text{reason}} + \alpha_c R^{(g)}_{\text{cons}}. \quad (4)$$

Table 1. Main results on naturalness (QualiSpeech), audio quality (NISQA), and emotion assessment (IEMOCAP). All LLM baselines except SALMONN-SQA are jointly finetuned on NISQA, QualiSpeech, and IEMOCAP.

	Naturalness				**Audio Quality**				**Emotion**				
Models	MSE↓	SRCC↑	PCC↑	FENSE↑	MSE↓	SRCC↑	PCC↑	FENSE↑	ACC↑	SRCC↑	CCC↑	PCC↑	FENSE↑
Non-LLM baselines													
Non-LLM	**0.602**	0.513	0.535	N.A.	0.538	0.739	0.786	N.A.	0.607	**0.628**	0.542	0.606	N.A.
LLM baselines													
SALMONN-SQA [3]	0.734	0.303	0.220	0.467	0.556	0.611	0.629	0.499	0.263	0.270	0.243	0.210	0.279
SALMONN-7B	0.741	0.457	0.421	0.571	1.866	0.680	0.676	0.331	0.368	0.212	0.221	0.225	0.248
Qwen2-Audio	2.590	0.047	0.067	0.442	0.935	0.489	0.474	0.662	0.229	0.406	0.412	0.422	0.412
Qwen2-Audio-Instruct	0.671	0.264	0.118	0.480	0.755	0.554	0.549	0.839	0.330	0.569	0.552	0.568	0.403
Ours													
D-UNIQE	0.622	**0.726**	**0.727**	**0.831**	**0.336**	**0.808**	**0.824**	**0.853**	**0.658**	0.606	**0.606**	**0.624**	**0.687**

The non-LLM baselines are UTMOS for naturalness, PESQ for audio quality, and Emotion2Vec for emotion assessment, respectively. Best results are in bold and second-best results are underlined. FENSE is reported only for models that generate natural-language descriptions.

The outcome reward is a bounded score-accuracy term

$$R_{\text{out}}^{(g)} = \text{clip}\Big(1 - \tfrac{|\hat{s}^{(g)} - s|}{\sigma_s},\, 0, 1\Big). \tag{5}$$

R_{fmt} rewards a task-appropriate response structure, and R_{reason} measures explanation quality, including acoustic grounding, perceptual coverage, and semantic similarity to the reference description. R_{cons} measures consistency between the predicted score and its textual explanation.

To prevent fluent but ungrounded explanations from dominating when the score is wrong, a quality-trust gate $\tau \in [0, 1]$ is computed once per sampled group. Let $\bar{R}_{\text{good}}$ and $\bar{R}_{\text{bad}}$ be the mean reasoning rewards on accurate ($|\hat{s} - s| \le \delta_g$) and inaccurate ($|\hat{s} - s| \ge \delta_b$) samples. Then

$$\tau = \begin{cases} 1 & \text{if } \bar{R}_{\text{good}} \ge \bar{R}_{\text{bad}}, \\ \exp\big(\bar{R}_{\text{good}} - \bar{R}_{\text{bad}}\big) & \text{otherwise.} \end{cases} \tag{6}$$

When inaccurate predictions yield stronger reasoning scores, τ shrinks the reasoning contribution.

The group-relative advantage is $\hat{A}^{(g)} = (R^{(g)} - \mu_R)/(\sigma_R + \epsilon)$. Letting $r_{g,\ell} = \pi_\theta / \pi_{\theta_{\text{old}}}$ be the token-level importance ratio and $\bar{r}_{g,\ell} = \text{clip}(r_{g,\ell}, 1 \pm \epsilon_c)$ its clipped counterpart, the GRPO-QTR objective is

$$\mathcal{L}_{\text{GRPO}} = -\frac{1}{G} \sum_{g=1}^{G} \frac{1}{|y^{(g)}|} \sum_{\ell} \min\Big(r_{g,\ell} \hat{A}^{(g)},\, \bar{r}_{g,\ell} \hat{A}^{(g)}\Big), \tag{7}$$

where $1/|y^{(g)}|$ prevents longer explanations from dominating the policy-gradient updates.

3.4. Training Objective

Training proceeds in two stages. In Stage 1, the model is trained with supervised language modeling together with the routing regularization:

$$\mathcal{L}_{\text{sup}} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{route}}. \tag{8}$$

Starting from the supervised checkpoint, Stage 2 applies GRPO-QTR to the score-based SQA tasks while retaining the supervised objective:

$$\mathcal{L} = \lambda_{\text{lm}} \mathcal{L}_{\text{sup}} + \lambda_{\text{rl}} \mathcal{L}_{\text{GRPO}} \tag{9}$$

Here, $\mathcal{L}_{\text{GRPO}}$ directly optimizes score accuracy and score–explanation consistency. GRPO-QTR is applied only to naturalness and audio-quality batches; emotion batches continue to use $\mathcal{L}_{\text{sup}}$.

4. EXPERIMENTS

4.1. Datasets

We train and evaluate on three public corpora: NISQA [19] for audio quality, QualiSpeech [8] for naturalness, and IEMOCAP [20] for emotion. For NISQA, noisiness, coloration, discontinuity, and overall MOS are converted into natural-language descriptions following the template protocol of AuditoryLLM [3]. QualiSpeech provides naturalness scores with perceptual descriptions covering human-likeness, speaking rate, and artifacts. IEMOCAP provides categorical emotion labels and dimensional ratings of valence, arousal, and dominance. The train/validation/test splits are $11{,}020/2{,}700/472$ for NISQA, $6{,}942/1{,}378/300$ for QualiSpeech, and $8{,}031/1{,}003/300$ for IEMOCAP.

4.2. Experimental Setup

Model We use Vicuna-7B-v1.5 with LoRA (rank 8, α=32). The frozen speech front-end consists of Whisper-large-v2 and BEATs, with a frozen w2v2-L-robust AVD encoder providing 1024-d affective features for the emotion task. Whisper features are modulated by ECAPA-TDNN-conditioned layer normalization. A windowed Q-Former with one learnable query token aggregates the task-dependent encoder features over 0.33 s windows.

Training We use AdamW with linear warm-up and cosine decay on a single A6000 GPU. Stage 1 jointly trains all tasks using supervised cross-entropy and the routing regularizer for 25 epochs (3,000 iterations per epoch, batch size 4, peak learning rate $5{\times}10^{-6}$). Stage 2 starts from the Stage 1 checkpoint and applies GRPO-QTR only to the two SQA tasks, mixed with supervised learning (λ_{lm}=0.9, λ_{rl}=0.1), using batch size 1 and 100 iterations per epoch. Emotion batches remain supervised with cross-entropy. We report the checkpoint with the best validation performance.

Instruction Design Each clip is evaluated with a task-specific instruction. For **naturalness**, the model predicts a 1–5 MOS-style score and explains human-likeness, prosody, speaking rate, and artifacts. For **audio quality**, it predicts a score with a short description

Table 2. Qualitative examples comparing ground-truth labels and D-UNIQE outputs on naturalness, audio quality, and emotion.

GT	D-UNIQE
Naturalness (QualiSpeech)	
The naturalness score is 4.0 out of 5.0. There are no distortions, and the audio flows very smoothly. The speech is mostly natural but slightly lacks full human-like quality. Overall, the quality is good.	The score is 4.0. There is no distortion, maintaining the natural sound of the speech. The audio flows very smoothly. The speech sounds mostly natural, closely resembling human speech, though it lacks some subtlety. Overall, the speech is rated as good.
Audio quality (NISQA)	
The score of the quality of the speech sample is 4.7. The speech sample is not distorted and fluent. The background noise is not noticeable. The overall quality of the speech sample is excellent.	The score is 4.2. The speech sample is not distorted and fluently continuous. The background noise is not noticeable. The overall quality of the speech sample is excellent.
Emotion (IEMOCAP)	
The speaker exhibits an angry emotion with a valence of 2.0, arousal level of 3.5 and dominance of 3.0. The speaker is male. Emotion scores include an anger score of 4.0 and frustration score of 3.0.	The speaker exhibits an angry emotion with a valence of 2.0, arousal level of 3.5 and dominance of 4.5. The speaker is male. Emotion scores include an anger score of 5.0.

of distortion, continuity, noise, and overall quality. For **emotion**, it predicts the expressed emotion together with valence, arousal, and dominance.

5. RESULTS

5.1. Evaluation

We evaluate three instruction-following tasks on held-out test clips: naturalness, audio quality, and emotion. For naturalness and audio quality, predicted scores are compared with human ratings using MSE, Spearman rank correlation (SRCC), and Pearson correlation (PCC). For emotion, we report categorical accuracy together with SRCC, concordance correlation (CCC), and PCC on valence, arousal, and dominance. For all LLM systems, we additionally evaluate generated descriptions with FENSE [21], which combines semantic similarity with a fluency-error penalty. FENSE is not applicable to non-LLM baselines, which produce scores only.

5.2. Results and Discussion

Table 1 summarizes the main results across naturalness, audio quality, and emotion assessment. Except for SALMONN-SQA, all LLM baselines are jointly finetuned on the same three corpora.

On naturalness, D-UNIQE achieves the best SRCC, PCC, and FENSE among the compared systems, while its MSE remains comparable to UTMOS. On audio quality, D-UNIQE achieves the best MSE, SRCC, and PCC, including against PESQ, and the highest FENSE among LLM systems. On emotion, it obtains the highest categorical accuracy, CCC, PCC, and FENSE, while remaining competitive with Emotion2Vec in SRCC.

Table 4 examines the contribution of individual components on matched clips. AVD conditioning improves emotion accuracy, dimensional affect agreement, and description quality, while its effect on SQA metrics is mixed. GRPO-QTR further improves naturalness and audio-quality correlation together with FENSE.

Table 3. Score–description consistency across projection strategies on matched n=200 naturalness clips.

Projection	FENSE† ↑	ρ ↑	$\mathrm{CR_H}$ ↓	$\mathrm{CR_L}$ ↓
Shared	0.558	<u>0.620</u>	5.5	14.3
Plain MoE	**0.709**	0.270	43.8	**0.5**
Soft task MoE	<u>0.688</u>	**0.890**	**0.0**	**0.5**

†Description-only FENSE with the leading score sentence removed. ρ denotes Spearman correlation between predicted score and description polarity. CR denotes contradiction rate. $\mathrm{CR_H} = P(\hat{s} \geq 4.5 \mid \mathrm{neg.})$ and $\mathrm{CR_L} = P(\hat{s} \leq 2.0 \mid \mathrm{pos.})$. Best results are in bold and second-best results are underlined.

Table 4. Component ablations of D-UNIQE on matched n=200 clips per task.

Setting	Emo. Acc.↑	AVD CCC↑	Nat. LCC↑	AQ LCC↑	FENSE↑
w/o AVD	0.550	0.517	**0.590**	0.811	0.562
w/ AVD	**0.642**	**0.603**	0.531	**0.835**	**0.610**
Stage 1	0.642	0.603	0.531	0.835	0.610
Stage 1 + GRPO-QTR	**0.658**	**0.606**	**0.547**	**0.845**	**0.790**

Bold indicates the better result within each ablation pair.

Table 3 evaluates score–description consistency across projection strategies. Although plain MoE achieves the highest description-only FENSE, it shows weak score–description agreement ($\rho = 0.270$) and frequent contradictions between high scores and negative descriptions ($\mathrm{CR_H}$). Soft task routing improves agreement to $\rho = 0.890$, eliminates the observed $\mathrm{CR_H}$ cases, and reduces low-score/positive-description contradictions from 14.3% to 0.5% ($\mathrm{CR_L}$). These results suggest that task-aware routing improves not only the fluency of generated descriptions but also their consistency with the predicted score.

Qualitative examples are provided in Table 2. For naturalness, D-UNIQE assigns a score of 4.0 and describes the speech as generally natural, while preserving the reference's qualification that it lacks some human-like subtlety. For audio quality, the generated explanation attributes the high score to low distortion, fluent continuity, and imperceptible background noise. For emotion, the model identifies the primary angry affect and produces dimensional affect estimates broadly consistent with the reference. These examples indicate that D-UNIQE generates score-conditioned descriptions grounded in task-relevant perceptual cues. A remaining limitation is that naturalness outputs still overuse audio-quality vocabulary, reflecting cue overlap in the QualiSpeech dataset rather than a failure of the intentional soft expert-sharing design.

6. CONCLUSION

We presented D-UNIQE, a unified framework for descriptive naturalness, audio-quality, and emotion assessment of expressive TTS. By combining a disentangled multi-encoder with a task-routed projection MoE and GRPO-QTR, D-UNIQE produces perceptually grounded, interpretable evaluations within a single model. Experiments show that it achieves competitive performance against specialist non-LLM baselines while generating more consistent natural-language outputs than jointly finetuned speech LLMs. Future work will further disentangle naturalness from audio-quality cues and refine the reward design to promote more faithful task-specific explanations and improve robustness in multi-task evaluation settings.

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